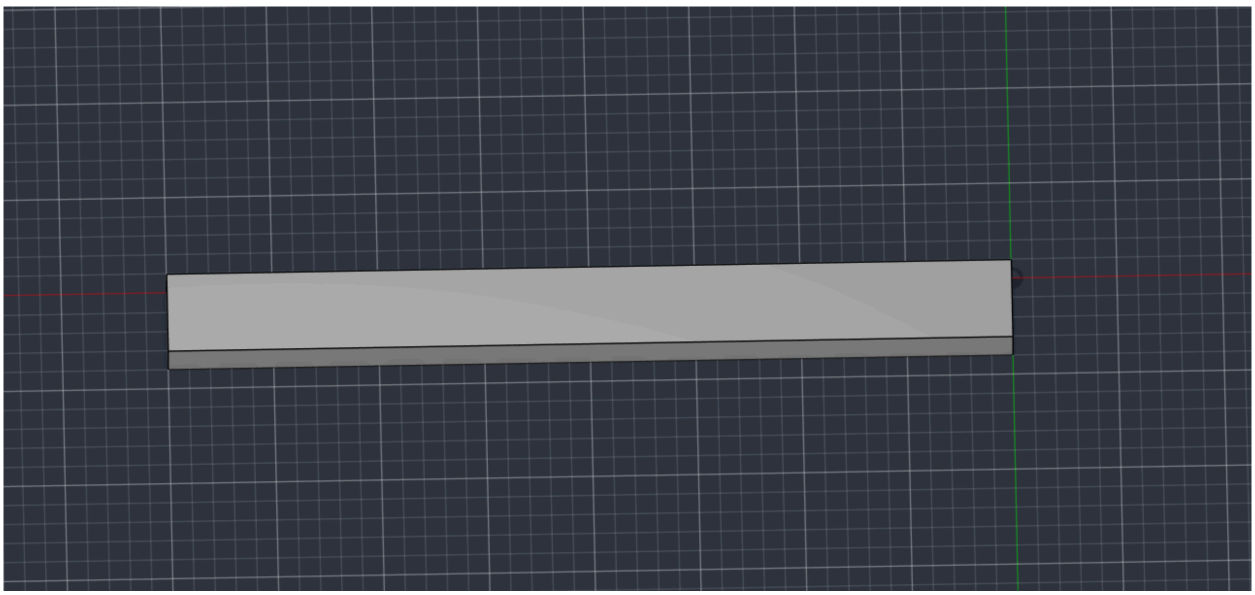
Cantilever Beam Stress & Deflection Analysis

Problem

Analyze a steel cantilever beam (200mm × 20mm × 10mm cross-section) fixed at one end with a 100N point load applied perpendicular to the beam at the free end. Goal: determine maximum bending stress and free-end deflection, and validate simulation results against classical beam theory.

Approach

Modeled the beam geometry in Autodesk Fusion 360 using parametric sketching and extrusion.



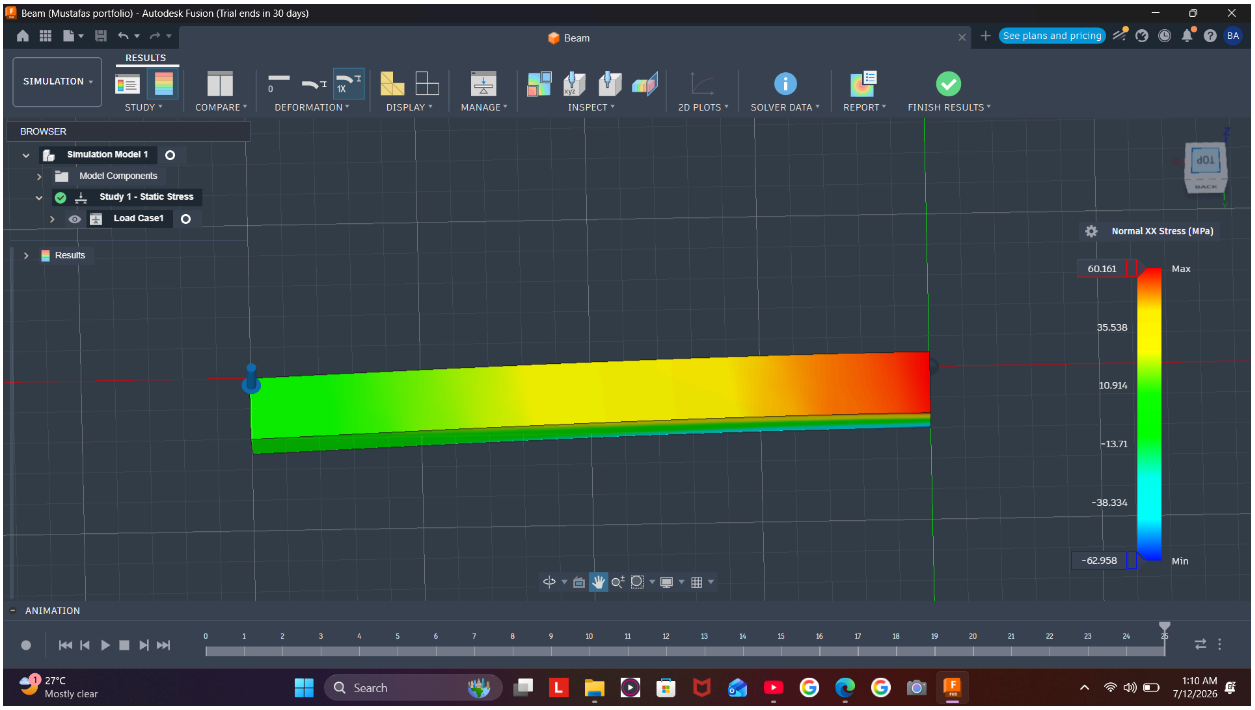
Applied steel material properties (E ≈ 200,000 MPa). Set up a Static Stress simulation study with a fixed constraint at one end and a 100N downward force at the free end. Generated a mesh and solved using Fusion's built-in FEA solver.

Hand Calculation

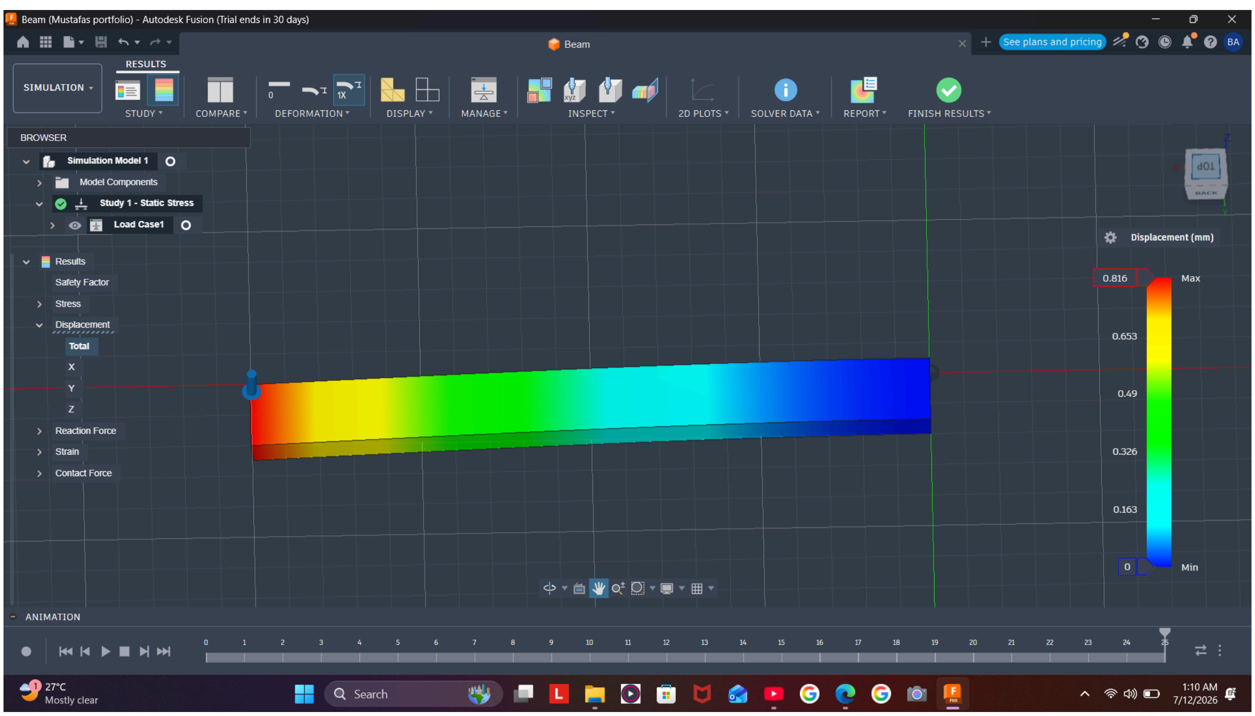
Using standard cantilever beam formulas ($\sigma = My/I$ for stress, $\delta = FL^3/3EI$ for deflection):

- Predicted max stress: ~60 MPa
- Predicted max deflection: ~0.8 mm

Simulation Results



- Von Mises stress: 53 MPa
- Normal (XX) stress near fixed end: 60.16 MPa



- Max deflection at free end: 0.816 mm

Validation
Normal stress matched hand-calculation within 0.3%, and deflection matched within 2% — confirming the simulation setup was physically accurate.

Key Learning — Stress Singularities
A concentrated point load applied to a single small face created an artificial stress spike right at the load point, rather than the theoretically expected maximum at the fixed end. This is a well-known FEA artifact rather than a real physical effect — recognizing it required checking stress values away from the load application point rather than trusting the highest colored value on the plot. In future models, applying loads as distributed pressure over a larger area would reduce this artifact.

What I'd Improve Next
I will try a distributed load instead of a point load, and test a more complex geometry (e.g., a bracket with a stress-concentrating hole) as project 2.